

Shyam Kishan

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Education

University of California, Santa Cruz

Expected June 2027

Bachelor of Science in Computer Engineering (GPA: 3.58 / 4.00)

Santa Cruz, California

- **Relevant Coursework:** Data Structures and Algorithms (C/C++), Principles of Computer Systems Design (C), Logic Design (Verilog), Computer Architecture (RISC-V)

Projects

Multithreaded HTTP/1.1 Server | C

- Engineered a high-performance multithreaded HTTP/1.1 server handling 5.8K+ requests/sec under 400 concurrent connections.
- Benchmarked thread pool sizing (1-16 threads) with wrk, identifying 2.4x throughput gains up to the system's 8-core limit before diminishing returns from thread over consumption.
- Reduced average latency from 132ms → 51ms (61% improvement) through parallel request handling.

Autonomous Obstacle-Avoiding Robot + Telemetry Web App | Arduino (C++), Web (HTML/CSS/JS), REST API (Python)

- Built a full-stack robotic control system using Flask, JavaScript, and Arduino, enabling real-time interaction between a web dashboard and physical hardware.
- Designed and implemented a REST API for robot control and telemetry, supporting commands, mode switching, and live sensor data streaming.
- Developed a serial communication interface in Python to parse custom binary sensor packets and ensure reliable data transmission from embedded hardware.
- Created a responsive frontend dashboard with real-time data visualization (Chart.js) to monitor robot state and control movement interactively.

Mitosis: FPGA-Based VGA Game | Verilog, Vivado, Basys3

- Designed and implemented a real-time VGA game on a Basys3 FPGA, rendering 8 moving molecules (4 red, 4 blue) that players sorted across a central "membrane" using hardware-level logic.
- Built a finite state machine (FSM) to control molecule behavior and membrane gating, ensuring correct signal timing and synchronization across VGA's pixel clock and refresh cycles.
- Debugged state transition and timing hazards to ensure the FSM produced correct outputs at precise clock cycles, preventing glitches in molecule rendering and collision detection.
- Applied digital logic design principles (clock domain management, synchronous state transitions) to build a responsive, hardware-driven game entirely in Verilog.

Client-Server Chat System | Python

- Built a TCP-based chat system with a rendezvous server and peer-to-peer client communication, implementing a custom text-based protocol inspired by HTTP.
- Designed and implemented client and server state machines (WAIT/CHAT modes), handling registration, peer discovery, half-duplex messaging, and grace connection tear-down.
- Implemented robust socket I/O and error handling, including malformed message detection, TCP connection life-cycle management, and clean shutdown on QUIT or connection loss.

Technical Skills

Languages: C, C++, Python, Verilog, RISC-V

Systems & Tools: Git, Linux/Unix, GCC, Valgrind, Docker, Vivado

Concepts: Systems Programming, Multithreading, Concurrency, TCP/IP, Socket Programming, Client-Server Architecture, Memory Management, Debugging, State Machines, Embedded Systems